EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 09

Basic Maths and Calculus

Circle, Ellipse, Exponential and Differentiation

By - Excellence Programme

14 September 2026

Topics to be covered

01 5-minute graph recap

02 Circle: equation, centre, radius and exact graph

03 Ellipse: standard equation, axes and exact graph

04 S.H.M. speed-position ellipse

05 Exponential functions and their graphs

06 Differentiation, turning points and quadratic practice

Graph-shape formula map

Modulus $y = |x|$: V-shape inverse graph $y = 1/x$: rectangular hyperbola

Parabola $y = x^2$ square-root graph $y = \sqrt{(x)}$

Kinetic energy $K \propto v^2$ pendulum period $T \propto \sqrt{(L)}$

Orbital period $T \propto R^{3/2}$ at fixed p, kinetic energy $K \propto 1/m$

How to draw any graph

1. Put the dependent quantity on the y-axis and independent quantity on the x-axis.

2. Check the domain (allowed values), intercepts, symmetry and turning points.

3. Mark enough exact points; draw the appropriate smooth curve or straight segments.

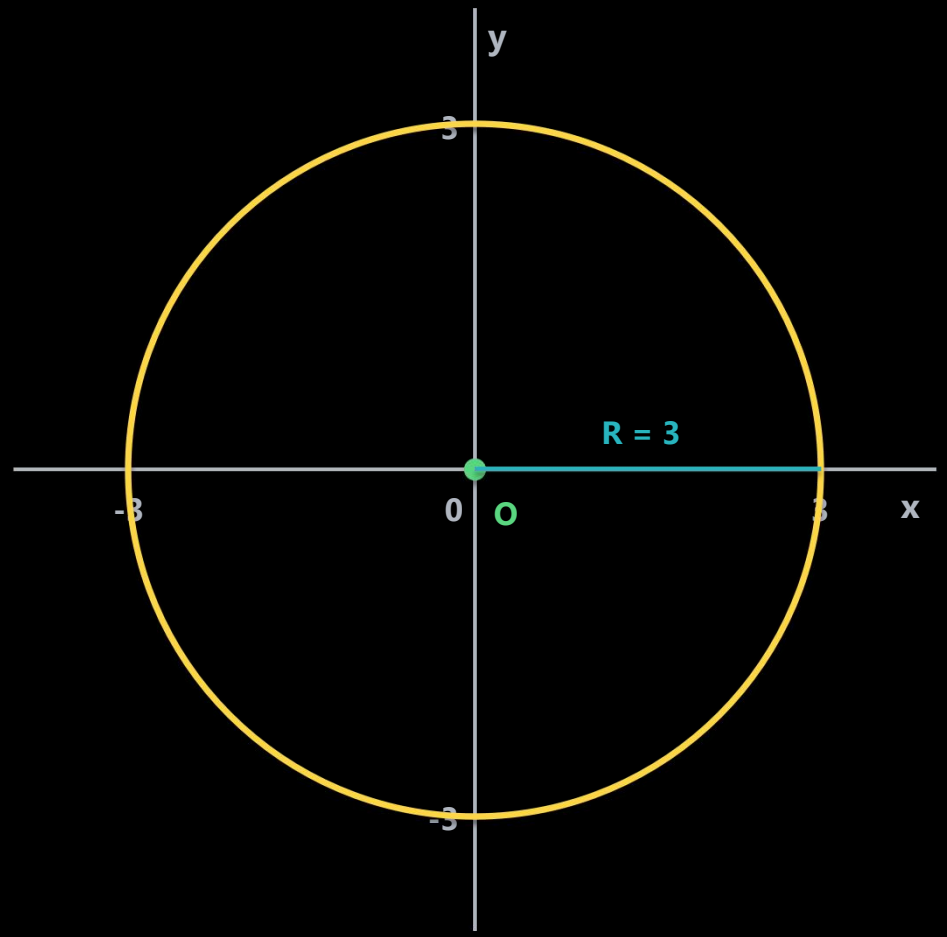
4. Label both axes and important points.

Circle: exact graph

A circle is the locus of points in a plane at a fixed distance R from a fixed point called the centre.

$$x^2 + y^2 = R^2$$

Shown here: R = 3 and centre = (0, 0).



Rough work / extra work

Equation of a circle: how it is formed

If the centre is C(h, k) and P(x, y) lies on the circle, then CP = R.

Distance formula: $\sqrt{[(x-h)^2+(y-k)^2]} = R$

$$(x-h)^2 + (y-k)^2 = R^2$$

centre = (h, k) **radius = R**

Signs inside brackets are opposite: $(x+3)^2$ means h = -3.

Important points on a circle

For $(x-h)^2+(y-k)^2 = R^2$:

right = (h+R, k) **left = (h-R, k)**

top = (h, k+R) **bottom = (h, k-R)**

These four points fix the exact width and height before drawing the circle.

Circle: $x^2+y^2 = 49$

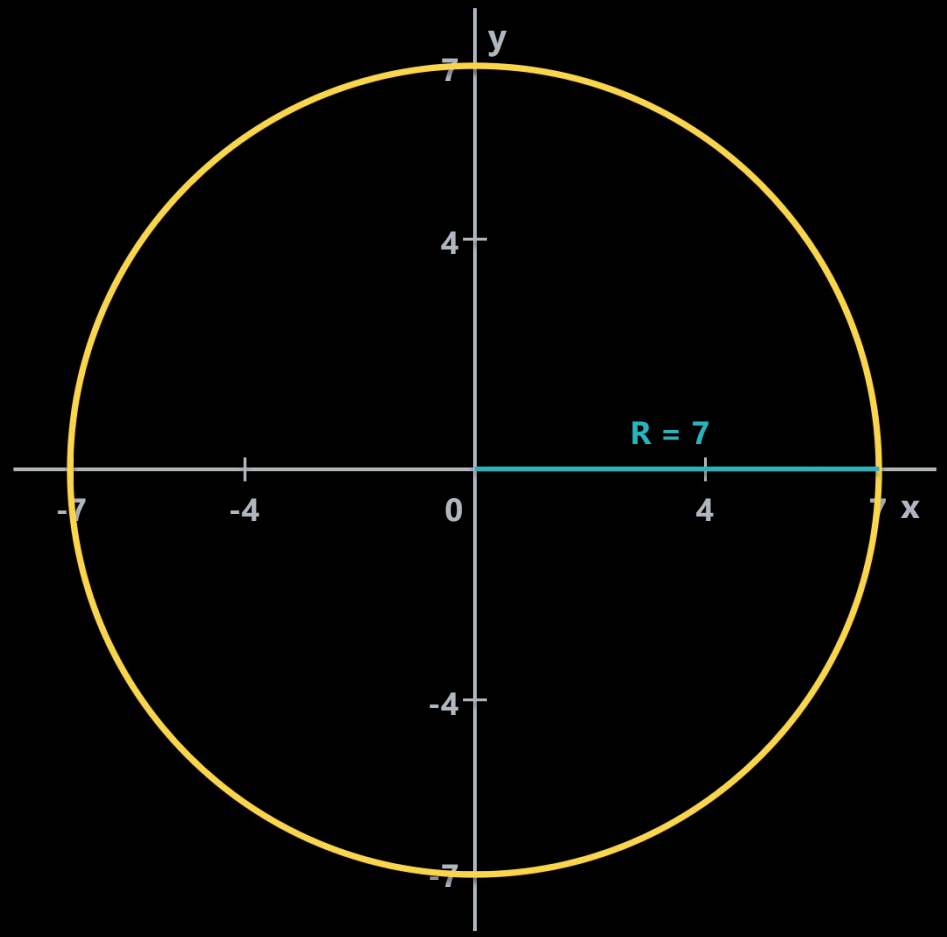
Find the centre and radius, then draw the exact graph.

Solution: Circle: $x^2+y^2 = 49$

Step 1: Compare with $x^2+y^2 = R^2$

Step 2: $R^2 = 49$, so $R = 7$

Centre = (0, 0), radius = 7.



Circle: shifted centre

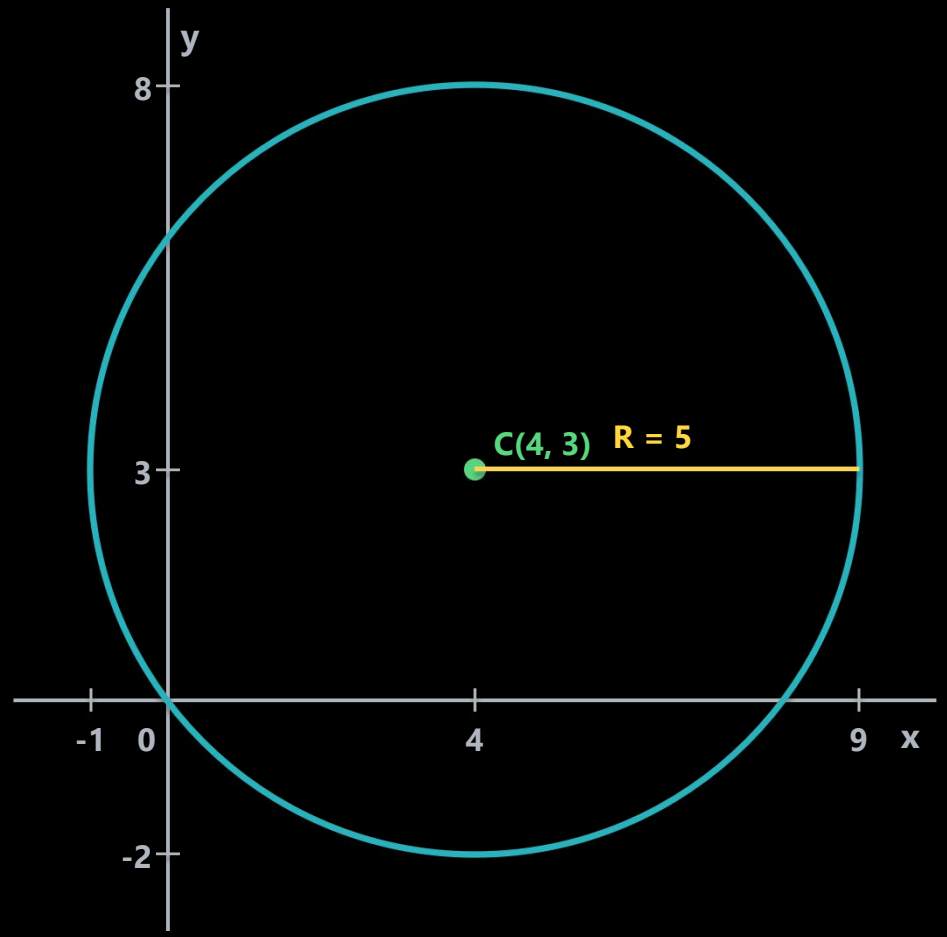
For $(x-4)^2+(y-3)^2 = 25$, find the centre and radius.

Solution: Circle: shifted centre

Step 1: Compare with $(x-h)^2+(y-k)^2 = R^2$

Step 2: $h = 4$, $k = 3$ and $R = \sqrt{(25)} = 5$

Centre = (4, 3), radius = 5.



Rough work / extra work

Circle: negative centre coordinate

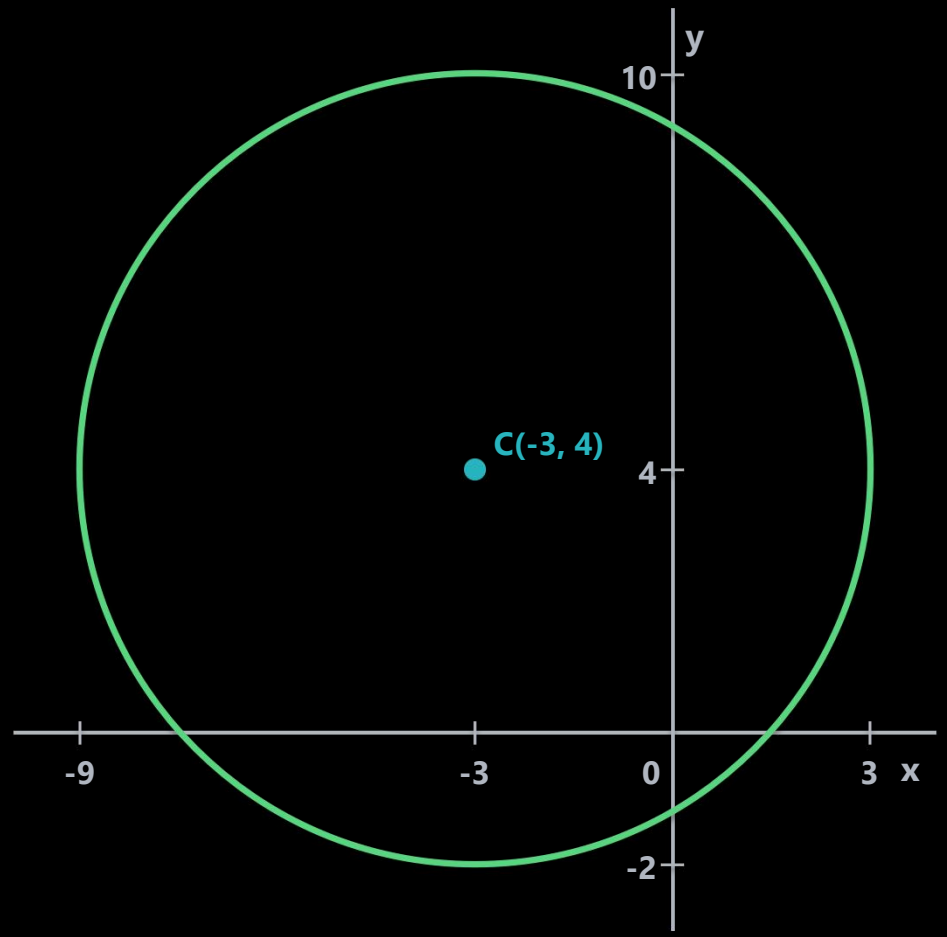
For $(x+3)^2+(y-4)^2 = 36$, find the centre and radius.

Solution: Circle: negative centre coordinate

Step 1: Write x+3 as x-(-3)

Step 2: h = -3, k = 4 and R = √(36) = 6

Centre = (-3, 4), radius = 6.



General equation of a circle

After expanding, a circle may appear as:

$$x^2+y^2+2gx+2fy+c = 0$$

centre = (-g, -f)

radius = $\sqrt{(g^2+f^2-c)}$

General circle equation

Find the centre and radius of $x^2+y^2-6x+4y-12 = 0$.

Solution: General circle equation

Step 1: $2g = -6$ gives $g = -3$; $2f = 4$ gives $f = 2$

Step 2: Centre = $(-g, -f) = (3, -2)$

Step 3: $R = \sqrt{[(-3)^2+2^2-(-12)]} = \sqrt{(25)} = 5$

Centre = (3, -2), radius = 5.

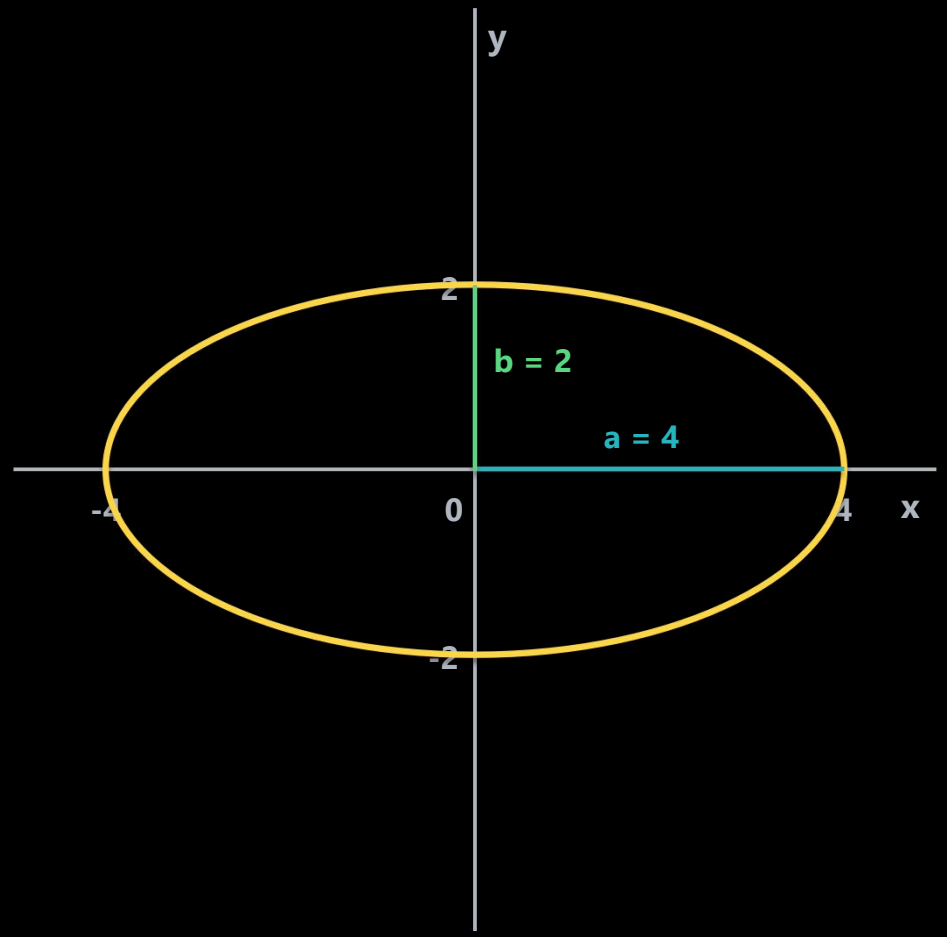
Rough work / extra work

Ellipse: exact graph

An ellipse is the locus of points whose total distance from two fixed points is constant.

$$x^2/a^2 + y^2/b^2 = 1$$

Shown here: a = 4 horizontally and b = 2 vertically.



Standard ellipse: how to read it

For an ellipse centred at the origin:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

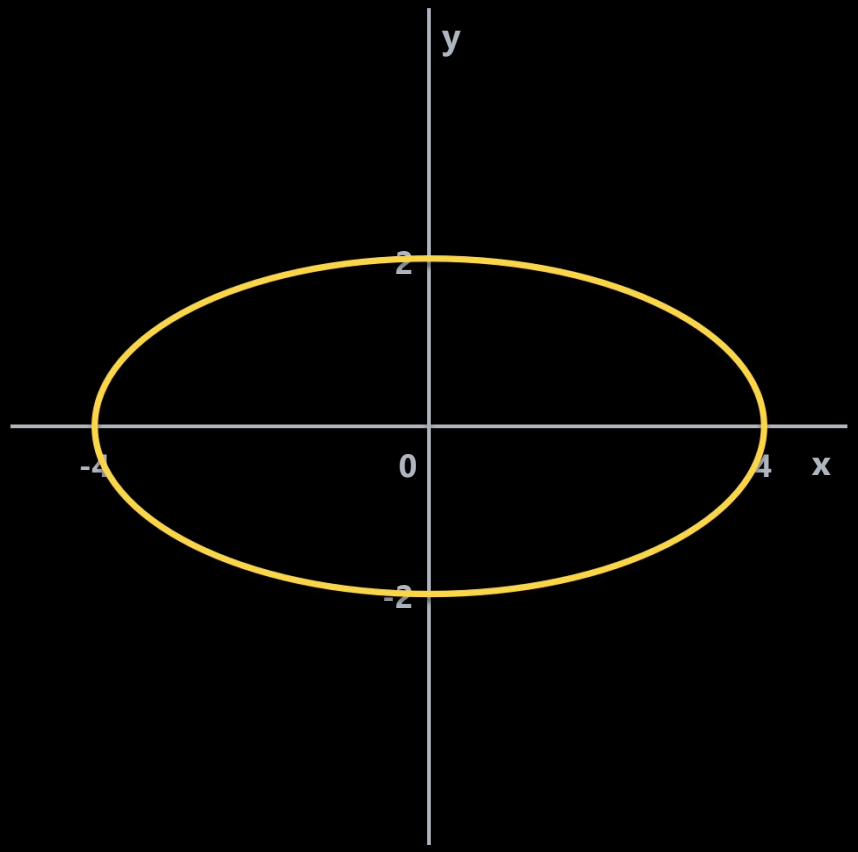
x-intercepts = (±a, 0)

y-intercepts = (0, ±b)

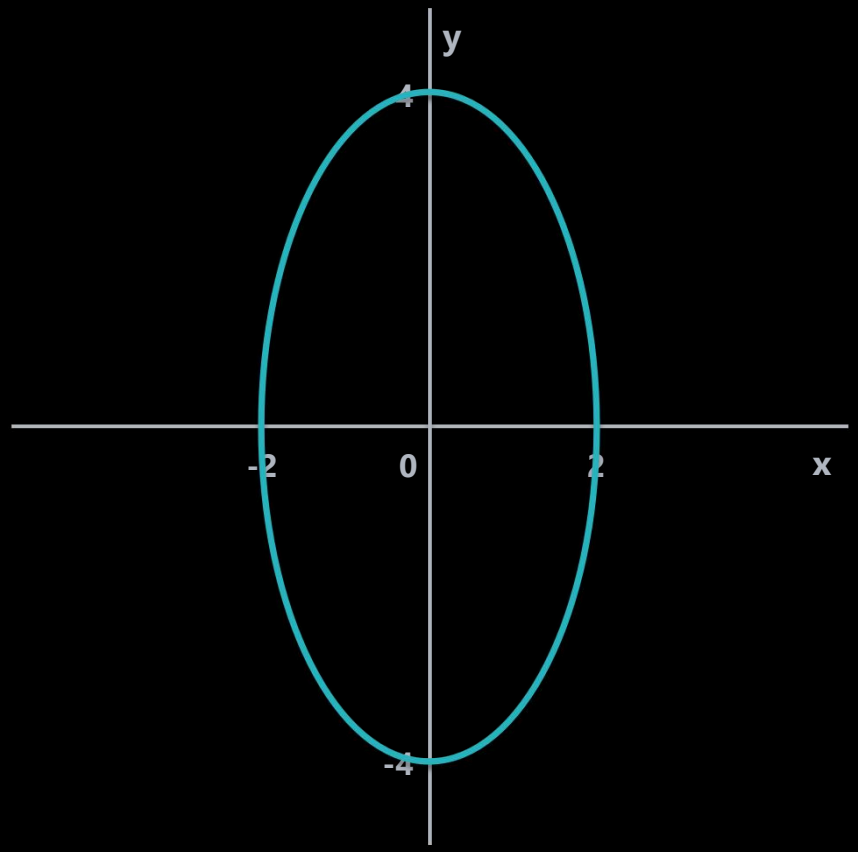
The larger denominator gives the longer semi-axis.

Horizontal and vertical ellipses

$a^2 > b^2$: horizontal major axis (a, b > 0)



$b^2 > a^2$: vertical major axis (a, b > 0)



Draw $x^2/4 + y^2/9 = 1$

Find the intercepts and draw the exact ellipse.

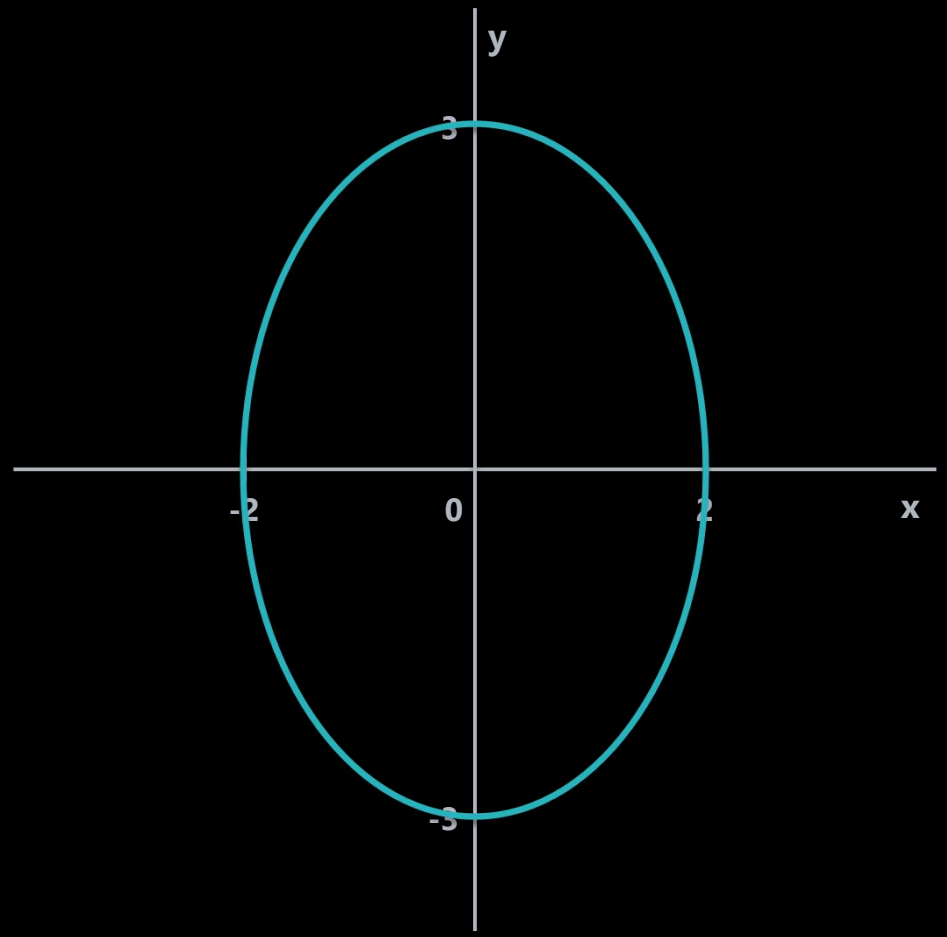
Solution: Draw $x^2/4 + y^2/9 = 1$

Step 1: $x^2/2^2 + y^2/3^2 = 1$

Step 2: x-intercepts = (+2, 0), (-2, 0)

Step 3: y-intercepts = (0, +3), (0, -3)

Vertical ellipse: semi-major axis = 3.



Rough work / extra work

Shifted ellipse

For $(x-2)^2/16 + (y+1)^2/4 = 1$, find centre and semi-axes.

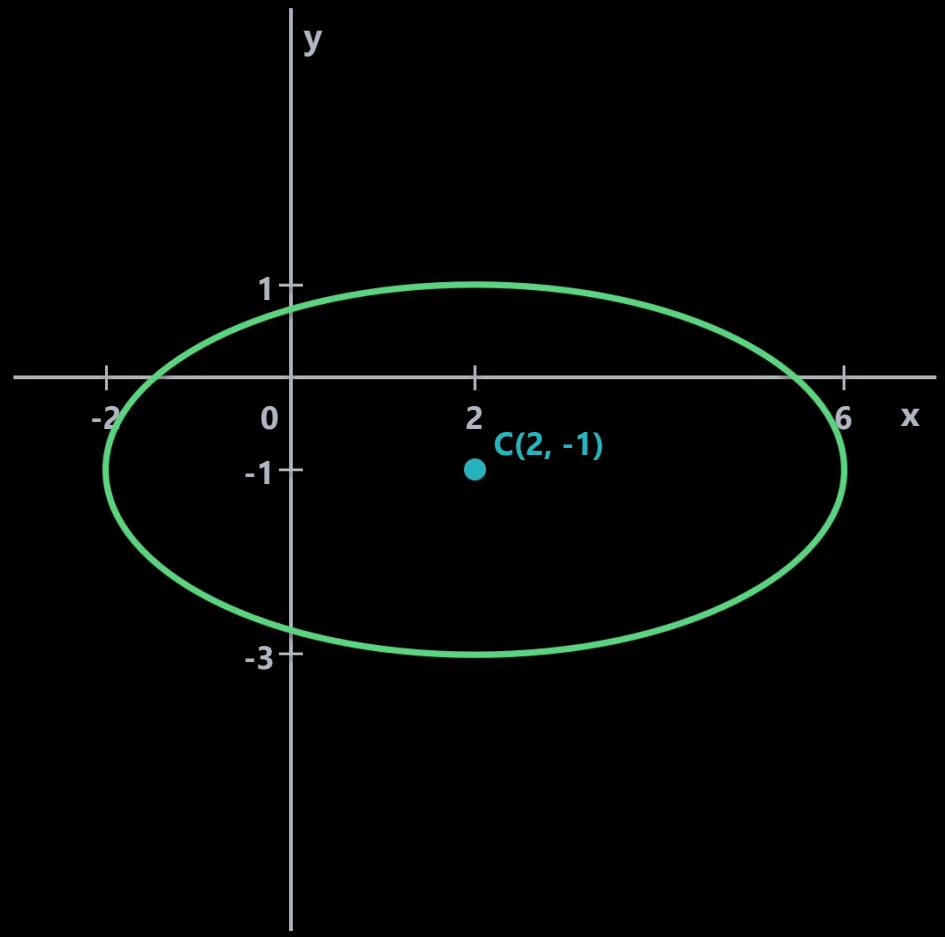
Solution: Shifted ellipse

Step 1: Centre = (2, -1)

Step 2: Horizontal semi-axis = √(16) = 4

Step 3: Vertical semi-axis = √(4) = 2

Horizontal ellipse centred at (2, -1).

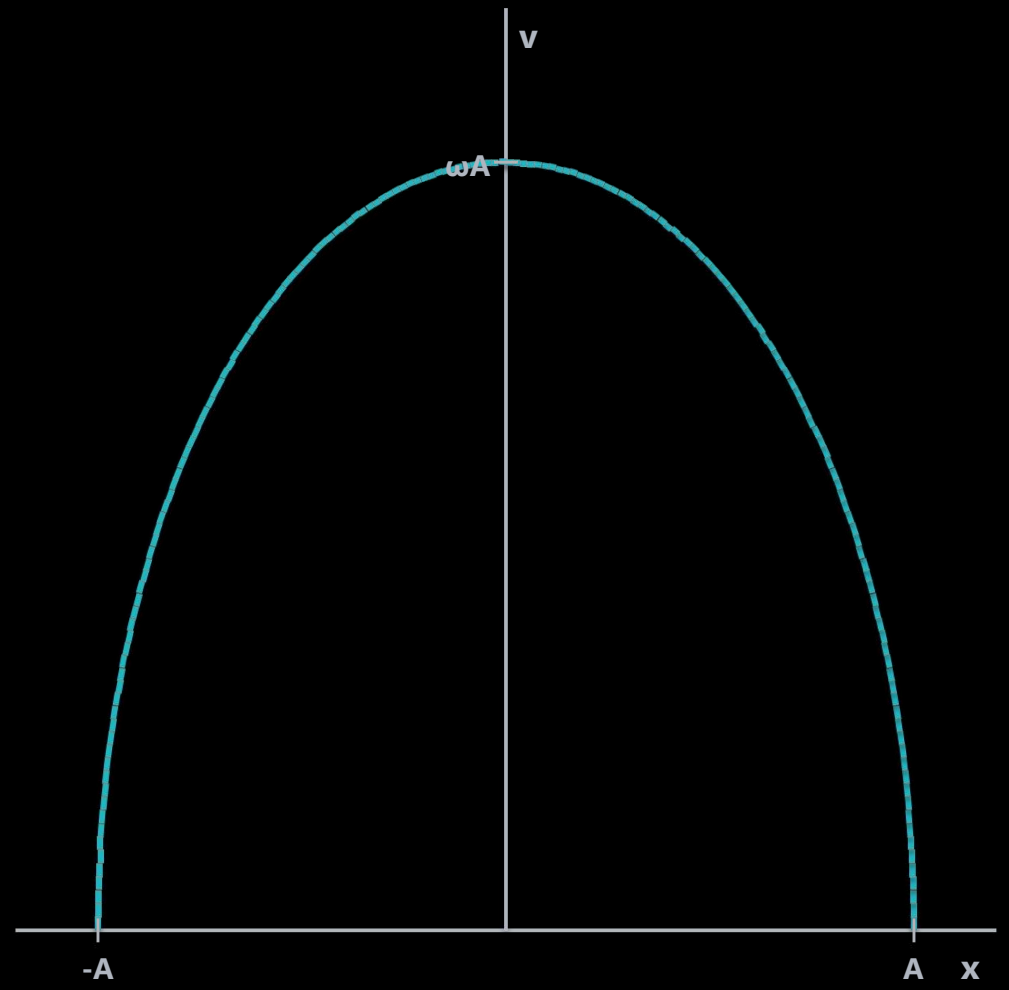


S.H.M. speed-position graph

For displacement x in simple harmonic motion:

$$|v| = \omega \sqrt{(A^2 - x^2)}$$

The speed-position graph is the upper half of an ellipse.



Theory of the S.H.M. ellipse

$v^2 = \omega^2(A^2-x^2)$

$x^2/A^2 + v^2/(\omega A)^2 = 1$

At x = 0, speed is maximum: $v_{max} = \omega A$.

At x = +A or -A, speed is zero because the particle turns back.

Velocity can be positive or negative; speed is only the upper half.

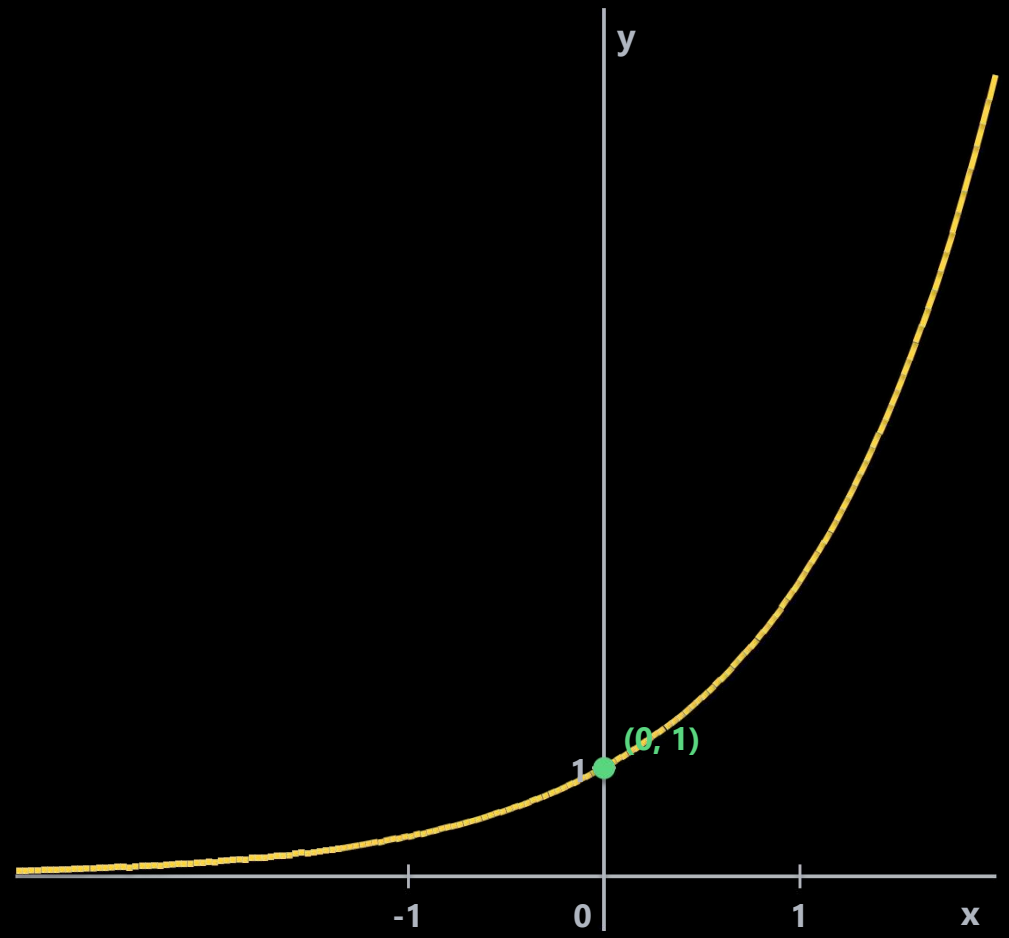
Rough work / extra work

Exponential graph: $y = e^x$

The variable x appears in the power.

e is approximately 2.718

The graph passes through (0, 1) and rises rapidly.



Properties of $y = e^x$

$e^0 = 1$, so y-intercept = (0, 1)

e^x is positive for every real x, so the graph never crosses the x-axis.

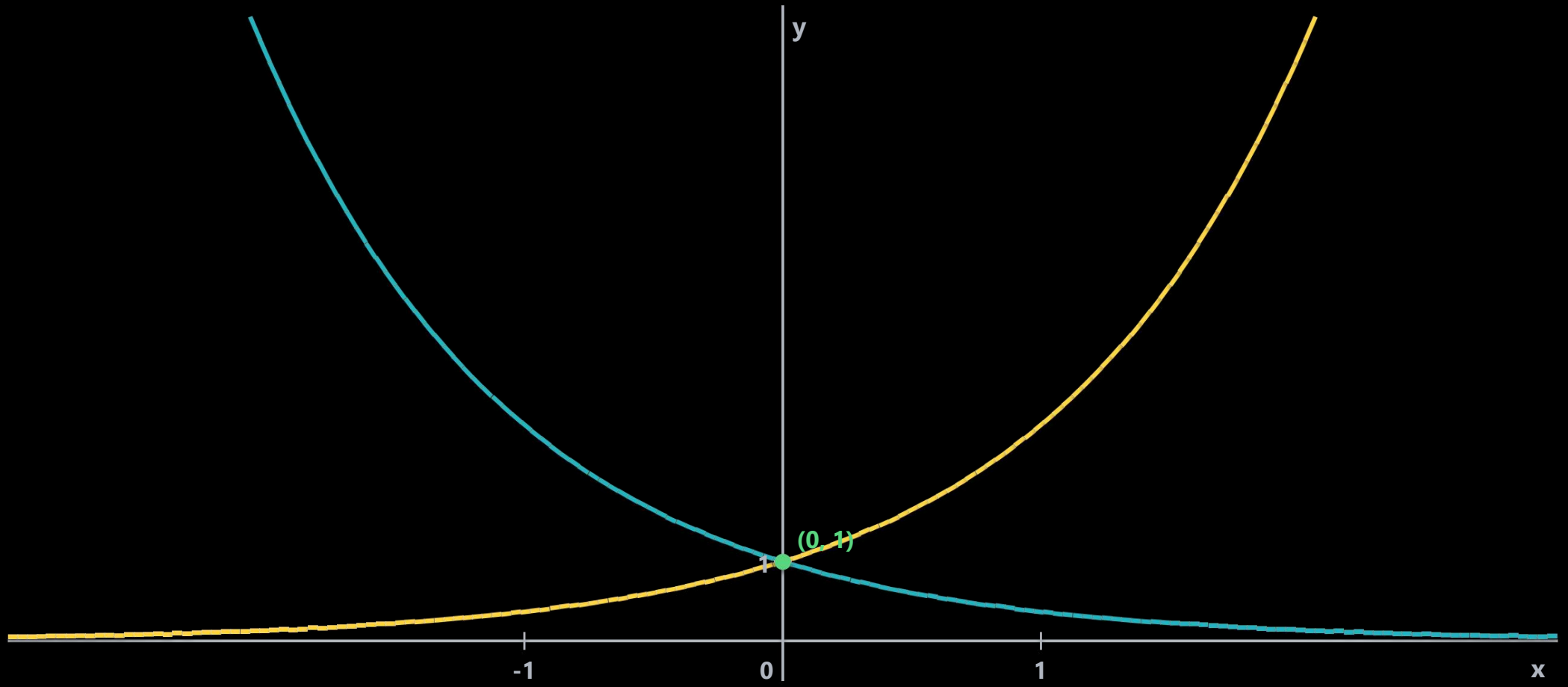
As x increases, e^x grows rapidly. As x becomes very negative, e^x approaches zero.

domain: all real x range: y > 0

Horizontal asymptote: y = 0.

$y = e^x$ and $y = e^{-x}$

Gold: e^x Teal: e^{-x}. Both pass through (0, 1).



Exponential values

For $y = e^x$, find y at x = 0, x = 1 and x = -1.

Solution: Exponential values

Step 1: $x = 0$: $y = e^0 = 1$

Step 2: $x = 1$: $y = e$

Step 3: $x = -1$: $y = e^{-1} = 1/e$

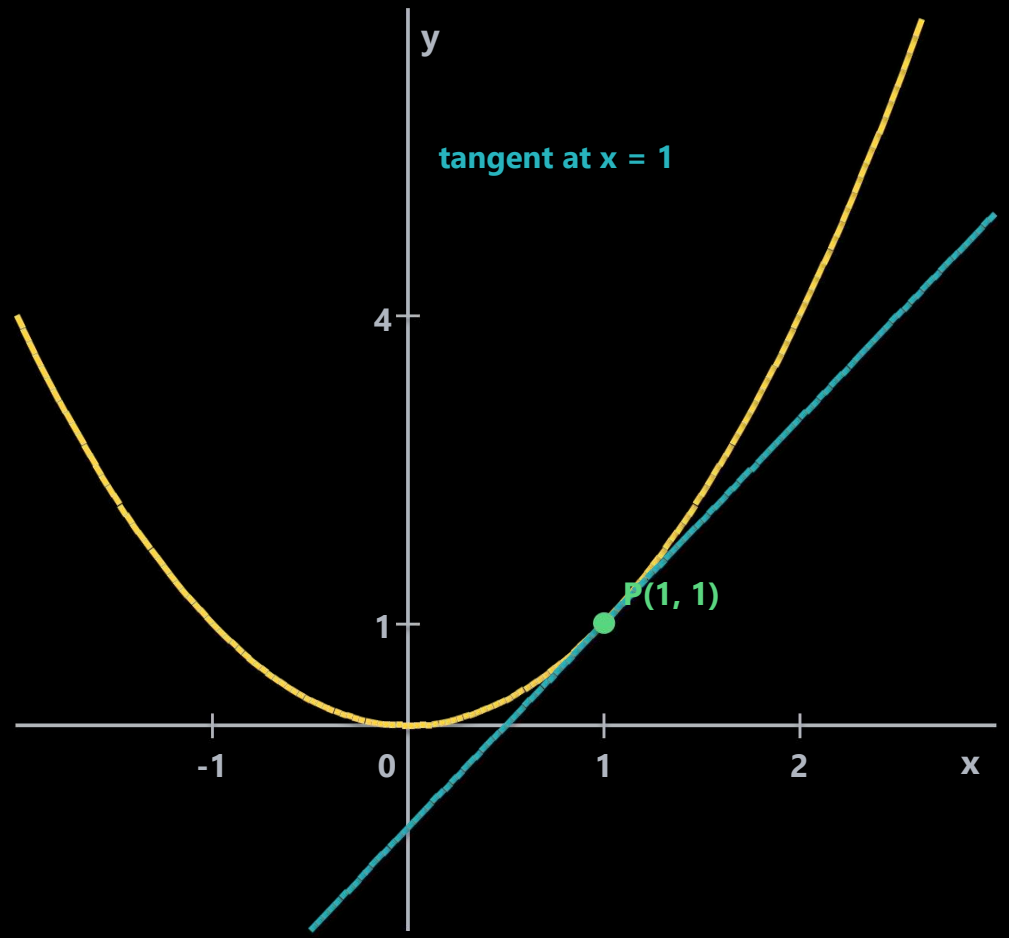
Points: $(0, 1)$, $(1, e)$, $(-1, 1/e)$.

Rough work / extra work

Differentiation: slope at a point

A curve has a different slope at different points.

The slope at one point is the slope of the tangent there.



How a derivative gives slope

Take two nearby points on a curve. Their joining line is a secant.

average slope = change in y / change in x

Move the second point closer to the first. The secant becomes the tangent.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$$

The derivative tells the instantaneous rate of change.

Ways to write a derivative

dy/dx $f'(x)$ y'

All three mean the derivative of y with respect to x.

units of derivative = units of y / units of x

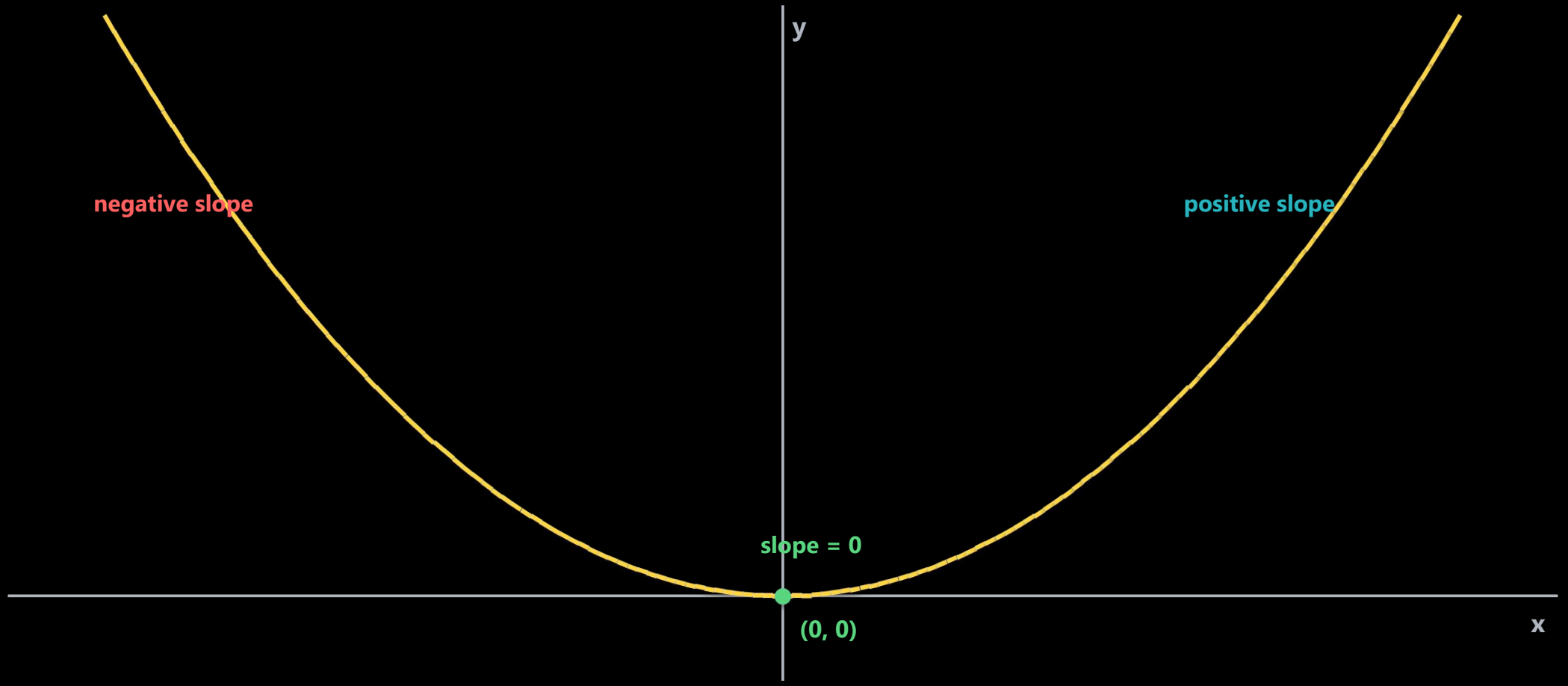
Example: slope of a position-time graph has unit metre per second (m s^{-1}) and represents velocity.

Sign of the derivative

curve behaviour	tangent slope	derivative
rising	positive	$dy/dx > 0$
falling	negative	$dy/dx < 0$
horizontal tangent	zero	$dy/dx = 0$

Positive, zero and negative slope

Gold curve: $y = x^2$. Left side falls, vertex has zero slope, right side rises.



Rough work / extra work

Maximum and minimum

At a smooth turning point, the tangent is horizontal.

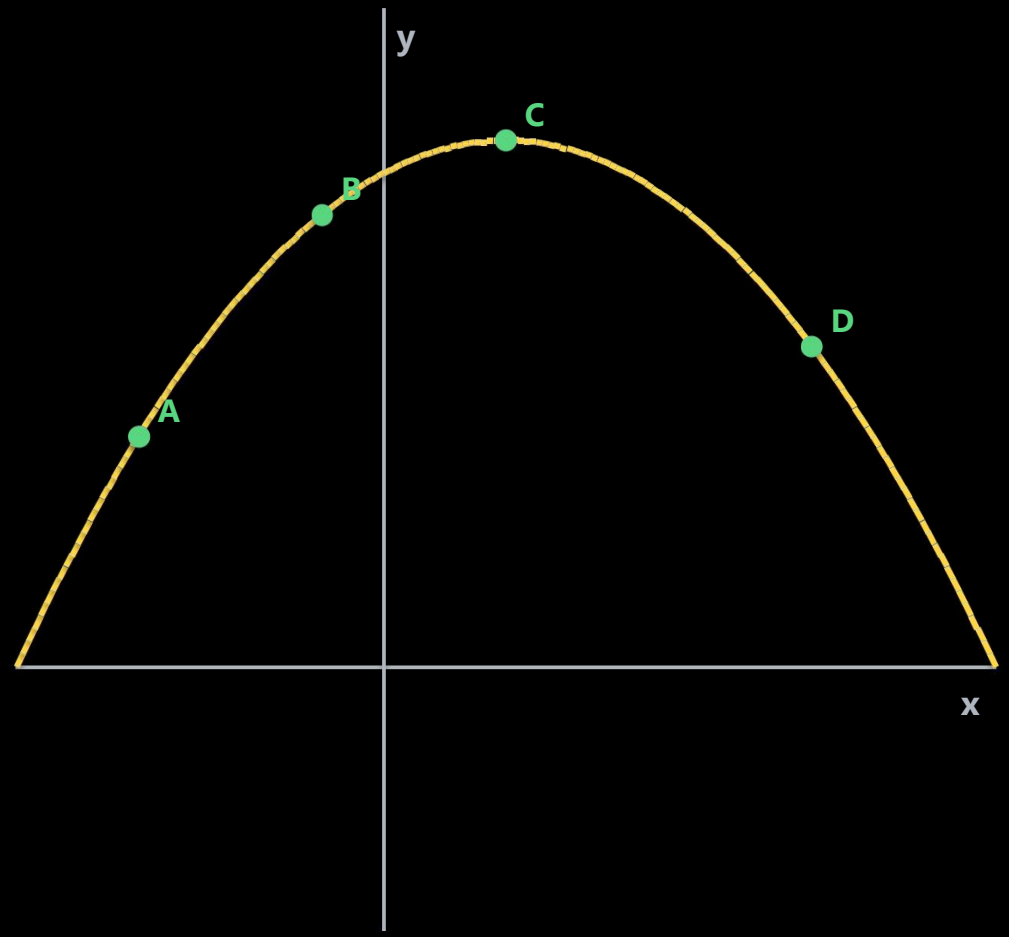
At a stationary point: dy/dx = 0

Minimum: derivative changes from negative to positive.

Maximum: derivative changes from positive to negative.

Match slope at points A, B, C and D

State the sign of the slope at A, B, C and D.



Solution: Match slope at points A, B, C and D

Step 1: A: positive slope

Step 2: B: positive slope

Step 3: C: zero slope because tangent is horizontal

Step 4: D: negative slope

A positive, B positive, C zero, D negative.

Power rule for differentiation

For constant n, where defined: if $y = x^n$, then $dy/dx = n\,x^{n-1}$

$d(x^2)/dx = 2x$

$d(x^3)/dx = 3x^2$

$d(\text{constant})/dx = 0$

Differentiate $y = x^2+3x-4$

Find dy/dx and the slope at x = 2.

Solution: Differentiate $y = x^2+3x-4$

Step 1: $d(x^2)/dx = 2x$

Step 2: $d(3x)/dx = 3$ and $d(-4)/dx = 0$

Step 3: $dy/dx = 2x+3$

Step 4: At $x = 2$: slope $= 2(2)+3 = 7$

Slope at $x = 2$ is 7.

Rough work / extra work

Quadratic roots: $x^2+7x+12 = 0$

Find both roots by factorisation.

Solution: Quadratic roots: $x^2+7x+12 = 0$

Step 1: Find two numbers with product 12 and sum 7: 3 and 4

Step 2: $x^2+7x+12 = (x+3)(x+4)$

Step 3: $(x+3)(x+4) = 0$

$x = -3$ or $x = -4$.

Quadratic roots: $2x^2+3x-2 = 0$

Find both roots by factorisation.

Solution: Quadratic roots: $2x^2+3x-2 = 0$

Step 1: Split 3x as 4x-x

Step 2: $2x^2+4x-x-2 = 0$

Step 3: $(2x-1)(x+2) = 0$

$x = 1/2$ or $x = -2$.

Rough work / extra work

Circle, ellipse and exponential

circle: $(x-h)^2+(y-k)^2 = R^2$

origin-centred ellipse: $x^2/a^2+y^2/b^2 = 1$

e^x passes through (0, 1), stays positive and has asymptote $y = 0$

Differentiation lock lines

The derivative gives the tangent's slope and represents an instantaneous rate of change.

rising: $dy/dx > 0$ falling: $dy/dx < 0$

stationary point: $dy/dx = 0$

$d(x^n)/dx = n\,x^{n-1}$

Practice

Redraw each circle, ellipse and exponential graph from this lecture with labelled axes.

Redo every question without looking at its answer slide.

Learn the meaning and sign of a derivative before the next class.